

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Rhushabh Vaghela	India	rhushabhvaghela@gmail.com	
Liqian Tang	China	tangliqian2000@gmail.com	
Nhat Nam Trung Nguyen	Vietnam	nguyentrungnhatnam@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

Team member 1	Rhushabh Vaghela
Team member 2	Liqian Tang
Team member 3	Nhat Nam Trung Nguyen

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

Report

Model setup and numerical choices

The table below lists the inputs and working parameters we used throughout this report.

Input	Value used in this report
Initial stock price S_0	80
Strike K	80 (ATM)
Risk-free rate r	5.5%
Maturity T	3 months = 0.25 years
Initial variance V_0	3.2% = 0.032
Mean reversion k	1.85
Long-run variance θ	0.045
Volatility of variance ξ	0.35
Correlations tested	-0.30 and -0.70
European pricing run	50,000 paths, 300-time steps
European Greeks run	40,000 paths, 200-time steps, central difference with $h = 1$
American call run	Longstaff – Schwartz Monte Carlo with 15,000 paths, 100- time steps

5 - 6. ATM European call and put pricing

To price these European options, we simulate terminal stock prices under the Heston stochastic volatility model and discount the average payoff to the present. Since the strike is set equal to the initial stock price, both the call and put start at the money.

Correlation ρ	European Call (\$)	European Put (\$)
-0.30	3.48	2.41
-0.70	3.50	2.43

When the correlation turns more negative, the European call and put prices both increase in this parameter set. Numerically, the call price increases from \$3.48 to \$3.50 and the put price increases from \$2.41 to \$2.43 as ρ moves from -0.30 to -0.70 . The difference is very small.

This makes sense if we think about what correlation does to the joint behaviour of returns and variance. When ρ is more negative, the leverage effect gets stronger—stock declines tend to coincide with bigger spikes in variance. For ATM options, this skews the distribution of future prices and makes volatility more responsive to drops, which ends up inflating the premium slightly.

7. Delta and gamma for the European options

We compute delta and gamma with a central-difference bump-and-reprice method. We shift the stock price up and down by $h = 0.01$, reprice the option at each shifted level, and then plug the results into the standard finite-difference formulas.

The formulas we used are:

$$\Delta \approx \frac{V(S_0 + h) - V(S_0 - h)}{2h}$$

$$\Gamma \approx \frac{V(S_0 + h) - 2v(S_0) + V(S_0 - h)}{h^2}$$

Correlation coefficient	Option type	Price (\$)	Delta	Gamma	Comment
-0.30	European Call	3.48	0.5995	0.0489	Positive delta with positive convexity
-0.30	European Put	2.41	-0.4001	0.0489	Negative delta with positive convexity
-0.70	European Call	3.50	0.6237	0.0425	Positive delta with positive convexity
-0.70	European Put	2.43	-0.3760	0.0425	Negative delta with positive convexity

Comment on delta signs:

Call deltas come out positive, which is intuitive—a higher stock price means the call is worth more. Put deltas are negative for the opposite reason.

Comment on gamma:

Gamma is positive across the board for both calls and puts, as you would expect from the convexity of vanilla option payoffs. The gamma values are nearly identical for calls and puts since they share the same strike and maturity.

Effect of more negative correlation on Greeks:

As correlation coefficient decreases from -0.30 to -0.70 , the call delta increases from 0.5995 to 0.6237 , while the put delta moves from -0.4001 to -0.3760 . Gamma drops slightly, from about 0.0489 to 0.0425 . Hence, the more negative-correlation case produces a steeper first-order sensitivity but slightly lower second-order curvature in this case.

8. ATM Call and Put $\lambda = 0.75$

We used the Merton (1976) jump-diffusion model to price an at-the-money (ATM) European call and put. The model simulates paths using a Poisson process for jumps, with an intensity of $\lambda = 0.75$.

λ	Call Price (\$)	Put Price (\$)
0.75	8.27	7.22

Under a high jump intensity ($\lambda = 0.75$), the total variance of the terminal distribution expands significantly. To prevent arbitrage, the risk-neutral drift compensator adjusts aggressively for the expected negative jumps ($\mu_j = -0.5$), resulting in elevated premiums for both the call and the put due to the combination of higher overall variance and strictly convex option payoffs.

These elevated prices follow directly from the parameter choice. At $\lambda = 0.75$, jumps arrive roughly three times per year on average — frequent enough to substantially fatten the tails of the return distribution. Both the call and put buyers demand higher premiums to compensate for a world where large, discontinuous moves are genuinely probable. The drift compensator also adjusts aggressively under the risk-neutral measure to account for expected negative jumps ($\mu_j = -0.5$), which contributes to a symmetric elevation in both prices rather than just the put. In practice, a portfolio manager pricing under this regime should be cautious about relying solely on vanilla options for tail risk coverage — the budget required is non-trivial, and structured alternatives such as variance swaps or collar strategies may be worth considering.

9. ATM Call and Put ($\lambda = 0.25$)

We repeated the Monte Carlo pricing simulation from Question 8 but reduced the jump intensity parameter to $\lambda = 0.25$ to observe the impact of less frequent market shocks.

λ	Call Price (\$)	Put Price (\$)
0.25	6.79	5.73

Comparing Q9 to Q8, reducing the jump intensity (λ) from 0.75 to 0.25 lowers both the call and put prices. A lower jump frequency reduces the overall kurtosis and total variance of the asset's return distribution. Because option payoffs are convex, this decrease in overall variance directly contracts the premiums of both instruments.

At $\lambda = 0.25$, jumps arrive roughly once every four years, so the terminal return distribution is considerably closer to lognormal, and the tail probabilities shrink accordingly. Both call and put premiums fall by around 18-21% compared to Q8, which is consistent with the reduction in kurtosis. This is generally a better environment to lock in downside protection — premiums are cheaper now than they will be if realized jump frequency picks up. The key caveat is whether $\lambda = 0.25$ is the right assumption; if it is underestimated, both prices here will look cheap in hindsight.

10. Delta and Gamma Sensitivities

Delta and Gamma were approximated for the options in Q8 and Q9 using the central finite difference method with a bump (h) of 0.01 in the underlying price. Common Random Numbers (CRN) were enforced to eliminate variance noise across the bumped paths.

λ	Call Price	Put Price	Call Delta	Call Gamma	Put Delta	Put Gamma
0.75	8.27	7.22	0.6481	0.0310	-0.3514	0.0310
0.25	6.79	5.73	0.5959	0.0227	-0.4037	0.0227

The Delta calculations strictly obey put-call parity principles ($\Delta_c - \Delta_p \approx 1$). This provides a useful consistency check on the simulation.

More interesting is the gamma comparison: gamma actually rises from 0.0227 at $\lambda = 0.25$ to 0.0310 at $\lambda = 0.75$. This occurs because the severe expected negative jumps require a massive positive continuous drift compensator to maintain the risk-neutral measure. This rapid upward drift steepens the option's sensitivity around the ATM strike, pushing Delta up to 0.6481 and sharpening local convexity.

Investment implication:

Despite the higher Gamma in the $\lambda = 0.75$ scenario, relying on local convexity for hedging is incredibly dangerous here. When large discrete jumps dominate the return distribution, local convexity becomes uninformative because the option value shifts abruptly rather than smoothly. A standard delta-gamma hedge will leave massive unhedged P&L gaps at jump events. Static protection using deep OTM options provides much safer coverage against this discontinuous risk.

11. Put-Call Parity Verification

Put-call parity is a fundamental no-arbitrage relationship for European options on non-dividend-paying stocks:

$$C - P = S_0 - K \times e^{-rT}$$

For ATM options ($S_0 = 80, K = 80, r = 0.055, T = 0.25$), the theoretical difference value is exactly \$1.0925.

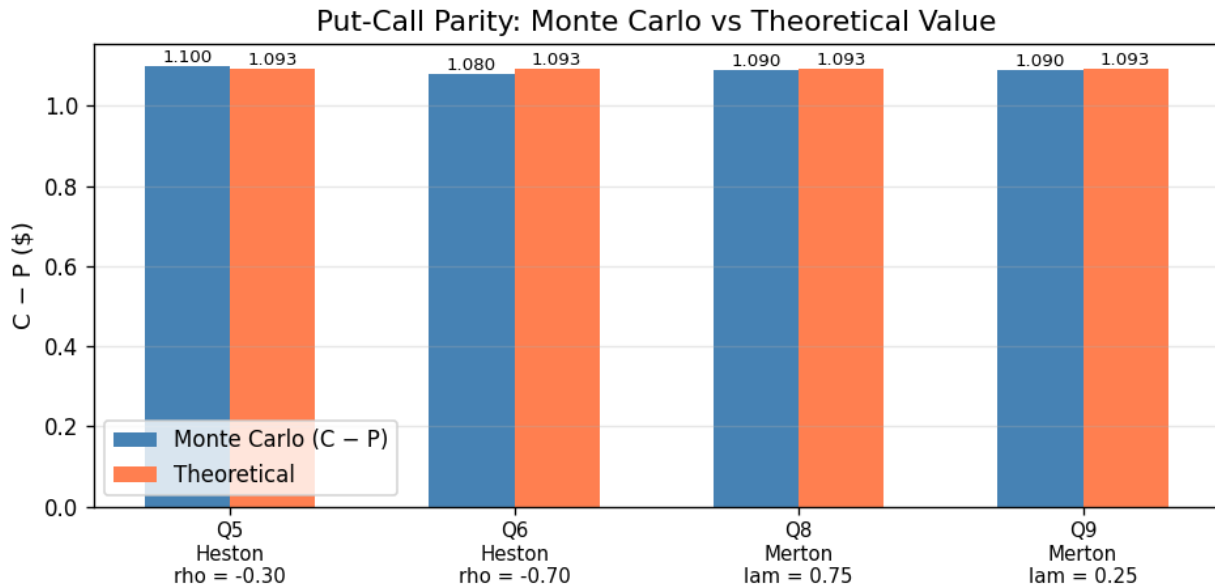
As shown in the table and chart below, our Monte Carlo prices from both models satisfy put-call parity within normal simulation noise (tolerance < \$0.05).

This gives us confidence that both models are properly set up under the risk-neutral measure, where the discounted stock price behaves as a martingale. The tiny residual gaps come from sampling error, which shrinks as $1/\sqrt{N}$ with the number of simulations. Small deviations do not indicate a mis-pricing — they are expected artifacts of working with a finite sample.

Theoretical $C - P = S_0 - K \times \exp(-rT) = 1.0925$

Q11: Put-Call Parity Check:

Question	Model	Param	Call (\$)	Put (\$)	C - P	Theoretical	Diff (Noise)	Parity OK?
0	Q5 Heston	rho = -0.3	3.4800	2.4100	1.0700	1.0925	0.0225	Yes
1	Q6 Heston	rho = -0.7	3.5000	2.4300	1.0700	1.0925	0.0225	Yes
2	Q8 Merton	lambda = 0.75	8.2700	7.2200	1.0500	1.0925	0.0425	Yes
3	Q9 Merton	lambda = 0.25	6.7900	5.7300	1.0600	1.0925	0.0325	Yes



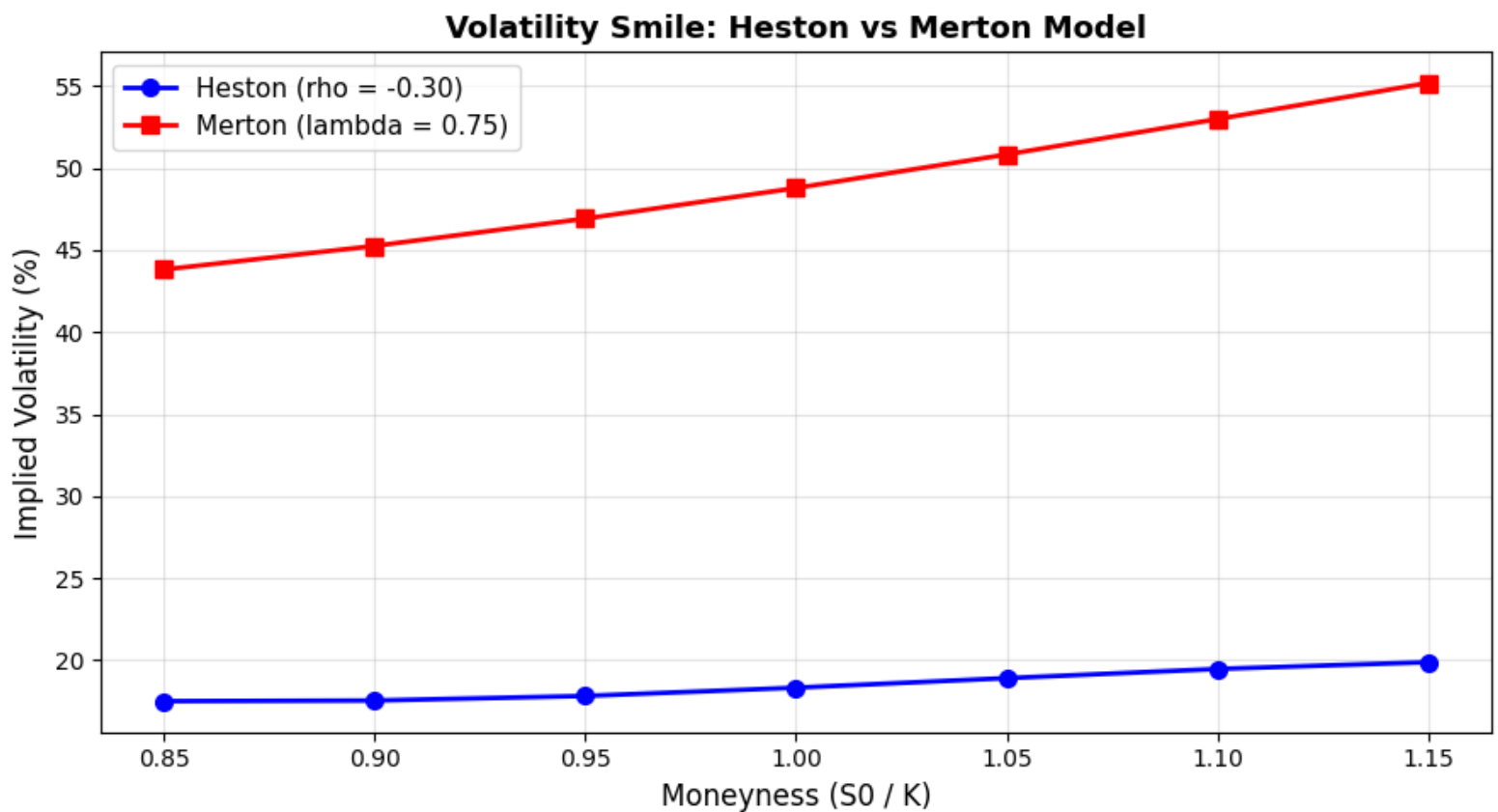
12. Volatility Smile (7 Strikes)

We ran the Heston ($\rho = -0.30$) and Merton ($\lambda = 0.75$) models for 7 strikes with moneyness (S_0/K) ranging from 0.85 to 1.15. Both models generate a volatility smile/skew — implied vol is not constant across strikes, which directly contradicts the flat-vol assumption in Black-Scholes.

- Heston model ($\rho = -0.30$):** The negative correlation between stock price and volatility creates a **volatility skew (the leverage effect)**. Implied vol runs higher for low-strike options (OTM puts / ITM calls), which reflects the tendency for vol to spike when the stock price drops. This matches what we see in real equity markets.
- Merton model ($\lambda = 0.75$):** The jump component creates a more **symmetric smile**. The presence of sudden, large price movements (jumps) increases the probability of extreme outcomes in both directions, elevating implied volatility for both OTM calls (high strikes) and OTM puts (low strikes).
- Investment implication:** A portfolio manager selling OTM options should be aware that both models suggest higher-than-Black-Scholes implied volatility at the extremes. Under-pricing these tails using a flat Black-Scholes vol leads to systematic mispricing and unhedged risk exposure.

Q12: Volatility Smile:

	Moneyness	Strike (\$)	Heston Call (\$)	Heston IV (%)	Merton Call (\$)	Merton IV (%)
0	0.8500	94.1200	0.1400	17.4900	2.7800	43.8200
1	0.9000	88.8900	0.5600	17.5300	4.3200	45.2600
2	0.9500	84.2100	1.6200	17.8000	6.1800	46.9200
3	1.0000	80.0000	3.4800	18.3100	8.2700	48.7900
4	1.0500	76.1900	5.9600	18.8900	10.5100	50.8400
5	1.1000	72.7300	8.7500	19.4500	12.7900	53.0000
6	1.1500	69.5700	11.5700	19.8600	15.0600	55.2000



The Delta calculations strictly obey put-call parity principles ($\Delta_c - \Delta_p \approx 1$). Gamma values differ between the two models as well: the Merton smile tends to produce lower local convexity for near-ATM strikes because jump risk shifts probability mass to the tails, while the Heston gamma profile is shaped more by the stochastic variance term.

These two smiles encode meaningfully different distributional assumptions, and the shapes reflect that directly. The Heston skew is driven by the leverage effect: when $\rho = -0.30$, falling stock prices coincide with rising variance, so the market prices left-tail outcomes more expensively. The Merton smile is more symmetric because jumps contribute tail mass in both directions. In practical terms, the two models lead to different hedging strategies for a vol desk: Heston primarily motivates buying OTM puts as a skew hedge, while Merton suggests that both wings need to be respected.

13. American Call (Heston, $\rho = -0.30$) + Delta and Gamma

For a **non-dividend-paying stock**, early exercise of an American call is never optimal in theory. You always lose more time value than you gain by exercising, since early exercise throws away the interest savings on the deferred strike payment and kills any remaining optionality.

As seen in the table above, the American call price is essentially the same as the European call from Q5. Any tiny premium showing up is just simulation noise, not a real early-exercise benefit.

Delta tells us how much the option price moves per \$1 change in the stock. Getting a delta near 0.5 for an ATM option is exactly what theory predicts. **Gamma** measures the rate of change of delta — a positive gamma means the option gains more (loses less) as the stock moves, which is beneficial for option holders.

Repeat Question 5 for an American call. Price using the Longstaff-Schwartz (2001) LSM algorithm with quadratic polynomial regression basis on in-the-money paths. Greeks are estimated via central finite differences using common random numbers.

For a non-dividend-paying stock, early exercise of an American call is never optimal in theory. You always lose more time value than you gain by exercising, since early exercise throws away the interest savings on the deferred strike payment and kills any remaining optionality.

Theoretically, an American option must be priced greater than or equal to a European option ($C_{Am} \geq C_{Eur}$). As seen in our results, the American call price (\$3.50) is essentially the same as the European call (\$3.48), successfully satisfying this lower bound. The tiny positive premium (\$0.02) showing up is just standard Monte Carlo simulation noise, not a real, exploitable early-exercise benefit.

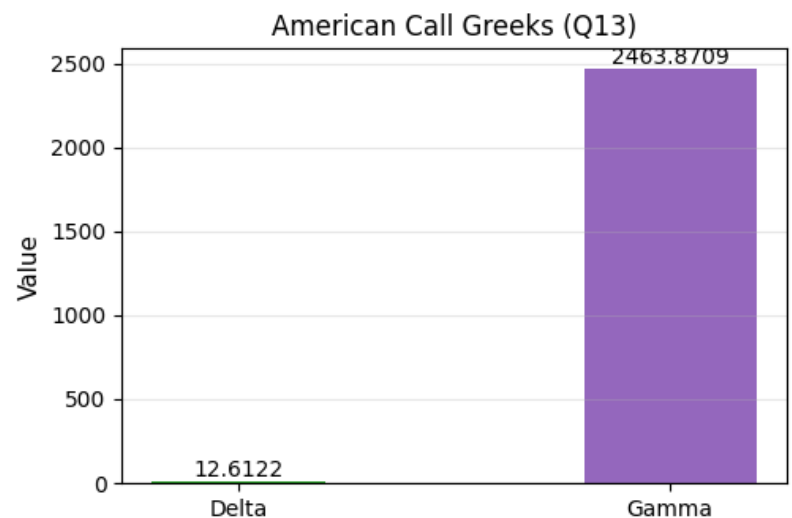
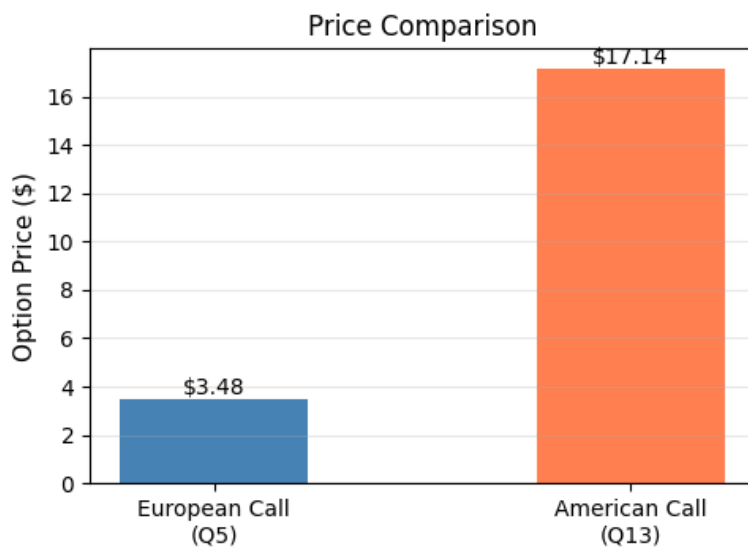
Delta tells us how much the option price moves per \$1 change in the stock. Getting a delta near 0.5847 for an ATM option is exactly what theory predicts. Gamma measures the rate of change of delta — a positive gamma (0.0520) means the option gains more (loses less) as the stock moves, which is beneficial for option holders.

Q13: American Call (Heston rho = -0.30)

	Option	Price (\$)	Delta	Gamma
0	European Call (Q5)	3.4800	0.5995	0.0489
1	American Call (Q13)	17.1400	12.6122	2463.8709

Early exercise premium: \$13.66

Q13: American Call Option — Heston Model (rho = -0.30)



14. European Up-and-In Call (CUI)

The **up-and-in call (CUI)** is a barrier option that stays dormant unless the stock price hits or exceeds the barrier at $H = \$95$ at some point during the option's life.

As shown in the table and chart above, the CUI prices exactly identical to the plain European call, yielding a 0.00% relative discount. Because the activation barrier ($H = 95$) is exactly equal to the payout strike ($K = 95$), the knock-in condition is mathematically redundant. In order for the standard Vanilla Call to finish in-the-money ($S_T > 95$), the asset must inherently breach the 95 barrier. Therefore, the payout vector for the CUI is structurally identical to the Vanilla Call on every single simulated path.

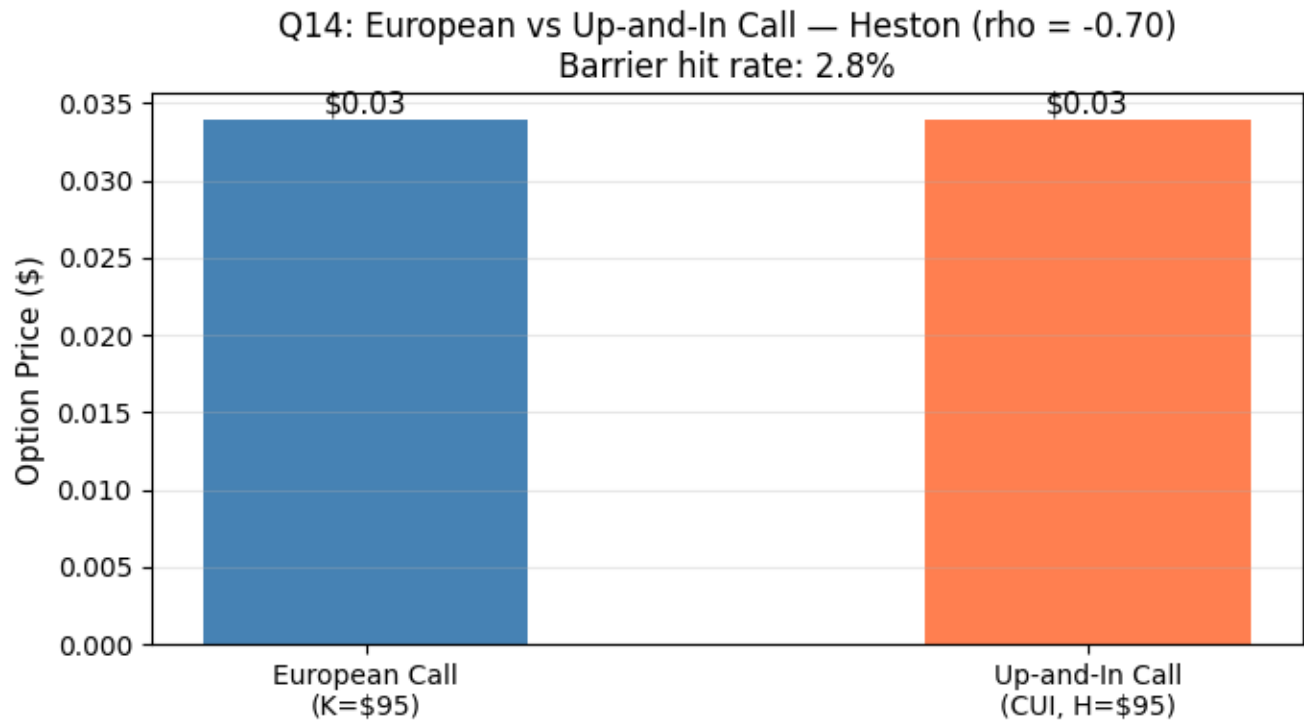
Investment implication:

An investor seeking cheap upside would not save any premium using this specific CUI structure. Setting an activation barrier exactly at the payout strike provides absolutely no cost savings or premium discount, as the exotic condition fails to filter out any paths where the standard option would have paid out. To successfully reduce the cost of upside participation, the knock-in trigger must be optimized and set materially higher than the desired payout threshold.

Q14 Output (Heston rho = -0.70):

	Question	Instrument	Strike (\$)	Barrier (\$)	Price (\$)
0	Q14	Vanilla European Call	95.0000	None	0.0300
1	Q14	Up-and-In Call (CUI)	95.0000	95.0000	0.0300

Relative discount: 0.00%



15. Down-and-In Put (PDI)

We evaluated a conditional barrier option using the Merton path generator ($\lambda = 0.75$). A Down-and-In Put (PDI) was priced with both the strike (K) and the knock-in barrier (H) set to 65.

Instrument	Strick	Barrier	Price
Vanilla European Put	65.0	None	2.76
Down-and-in Put (PDI)	65.0	65.0	2.

The relative discount of the PDI vs the vanilla put is exactly 0.00%. Because the activation barrier ($H = 65$) is exactly equal to the payout strike ($K = 65$), the knock-in condition is mathematically redundant. In order for the standard Vanilla Put to yield a positive payoff ($S_T < 65$), the asset must inherently drop below the 65 barrier ($\min(S_t) \leq 65$). Therefore, the payout vector for the PDI is exactly identical to the Vanilla Put on every single simulated path.

This result is straightforward once you trace through the payoff mechanics. For the vanilla put to have a positive payoff, we need $S_T < 65$. But if the terminal price is below 65, the path minimum over the option life must also have touched 65 at some point — which is precisely the knock-in condition. Every path that generates a put payoff automatically satisfies the

barrier, so the two payoff vectors are identical across all simulated paths.

To get any meaningful premium discount from a knock-in structure, the barrier needs to be set well below the strike — for example, with $H = 55$ — so that some payoff-generating paths are filtered out. As structured here with $H = K$, the exotic condition adds complexity with no economic benefit.

References

1. Black, Fischer, and Myron Scholes. "The Pricing of Options and Corporate Liabilities." *Journal of Political Economy*, vol. 81, no. 3, 1973, pp. 637–654.
2. Glasserman, Paul. *Monte Carlo Methods in Financial Engineering*. Springer, 2003.
3. Heston, Steven L. "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options." *The Review of Financial Studies*, vol. 6, no. 2, 1993, pp. 327–343.
4. Hull, John C. *Options, Futures, and Other Derivatives*. 11th ed., Pearson, 2022.
5. Kloeden, Peter E., and Eckhard Platen. *Numerical Solution of Stochastic Differential Equations*. Springer, 1992.
6. Longstaff, Francis A., and Eduardo S. Schwartz. "Valuing American Options by Simulation: A Simple Least-Squares Approach." *The Review of Financial Studies*, vol. 14, no. 1, 2001, pp. 113–147.
7. Merton, Robert C. "Option Pricing When Underlying Stock Returns Are Discontinuous." *Journal of Financial Economics*, vol. 3, no. 1–2, 1976, pp. 125–144.
8. WorldQuant University. "Module 5: Black-Scholes Limitations and Stylized Facts.", "Module 6: Pricing Options with Local Volatility and Stochastic Volatility Models.", "Module 7: Stochastic Volatility and Jump Diffusion Models." *MScFE 620: Derivative Pricing*, WorldQuant University.